X-Ray Diffraction Analysis of Silicon and Unknown Powder Sample

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Abstract

X-ray crystallography was performed on two samples: one containing a known silicon powder and one containing an unknown substance. A monochromatic X-ray beam was used, and a diffractogram was recorded by varying the detection angle. Comparing the results with a database and calculated theoretical values, the silicon sample exhibited peaks in intensity at the expected incidence angles. The calculated d-spacing and lattice constant matched the theoretical values closely. Using the same database, the unknown sample was suspected to be a mixture of KCl and NaCl.

1 Introduction

Scientists studied minerals for centuries before X-rays were discovered. By the late 1800s, mineralogists recognized that crystals had organized structures and speculated about atomic arrangements, but lacked direct evidence. The discovery of X-rays by Wilhelm Conrad Röntgen in 1895 advanced their research and understanding of crystal structures [1].

In the summer of 1912, 22-year-old Lawrence Bragg attended a lecture by Max Laue on X-ray diffraction, demonstrating that X-rays could diffract through crystals, confirming their wave nature [2]. Inspired, Lawrence and his father, William H. Bragg, researched at the University of Leeds. Lawrence discovered that diffraction patterns resulted from X-rays reflecting off atomic planes, leading to Bragg's Law. This breakthrough established X-ray crystallography, revolutionizing atomic structure analysis and earning multiple Nobel Prizes.

This report will use X-ray diffraction to analyze two powder samples; one known sample of silicon and one unknown sample. The diffraction patterns will be recorded digitally using an X-ray detector.

2 Theoretical background

Proposed by Bragg, his law is mathematically expressed as

$$n\lambda = 2d\sin\theta, \tag{1}$$

where n is the order of diffraction, λ is the wavelength of the incident beam, d is the interplanar spacing in a crystal and θ is the angle between the incident beam and the crystal plane. Bragg's law is a key principle in X-ray diffraction, explaining how X-rays interact with a crystal lattice to create constructive interference. This process produces diffraction patterns that reveal the material's structure. The geometric clarity and intuitive physical concepts behind Bragg's law make it a fundamental tool in the study of crystallography.

However, Bragg's formalism as an X-ray tool has certain limitations. The primary drawbacks we will address include its applicability only to infinite, perfect crystals, whereas diffraction requires fine grains. This requirement arises because the incident beam can penetrate only about 10^5 atomic layers [3]. If the sample is too large, the analysis will be limited to surface characteristics, which may be affected by imperfections such as impurities or oxidation layers. Furthermore, as seen from (1), Bragg's formalism does not provide information about diffraction intensities. We will systematically explore concepts that address these limitations.

In crystallography, the Miller indices $(h\,k\,l)$ are used to define specific planes within the crystal structure. They are integers that denote the orientation of a crystal plane relative to the crystallographic axes [3].

Working in the space of the reciprocal lattice, it is possible to derive certain relationships that will simplify our analysis of X-ray diffraction. For silicone, which crystallizes in a diamond cubic structure, it can be shown that the expression for the d-spacing, that is, the atomic plane distance, for the cubic unit cell, is [3]

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}, \tag{2}$$

where a is the lattice constant, and the integers h, k and l refer to the Miller indices of a given plane. If the crystallization is in the diamond structure, with an fcc Bravais lattice and a 2-atom basis with coordinates $(0, 0, 0)$ and $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, as is the case for silicone, then it is possible to show that the unit cell structure factor becomes [3]

$$F_{hkl} = f\left(1 + (-i)^{h+k+l}\right)\left(1 + (-1)^{k+l} + (-1)^{h+l} + (-1)^{h+k}\right),$$

where f is the form factor of the basis. Note that we have assumed a basis containing the same type of atoms. The allowed reflections of the crystal therefore occur whenever h, k and l are all even and $h + k + l = 4n$ for n an integer, or whenever h, k and l are all odd. The cubic structure factor will be important in pointing out the allowed Miller indices, and hence, the diffraction planes, in the crystal structure.

Knowing now the allowed reflections in the cubic crystal, it is easy to find the Miller indices of the first few reflecting planes. We consider, for instance, the Si structure with lattice constant $a = 5.4309$ Å and an incident beam of wavelength $\lambda = 0.7093$ Å [3]. Then, using (1) and (2) it is possible to find the first angular positions and the d-spacings corresponding to the allowed Miller indices.

3 Methods

3.1 Experimental setup

For the diffrection setup, a Mo-Kα X-ray source was used. In order to focus the beam around the characteristic wavelength $\lambda = 0.7093$ Å, and get rid of bremsstrahlung, a monochromatic filter was employed. The filter was produced by a graphite crystal, as opposed to a silicon crystal, trading some precision in the wavelengths produced for a higher photon flux. The diffraction setup was set to run at 40 kV and 30 mA for the whole analysis. The machine was set up to analyze samples from a start position of 1° to an end position of 40° over 391 steps, where each step had an exposure time of 1 second. The sample stage (see Figure 1) was rotational, rotating around its own axis, with a period of 1 second during the experiment. This was done to average out any differences in intensity caused by blemishes in the material. In order to shield critical equipment and the participants of the experiment from any harmful radiation, several lead plates were employed in the setup.

3.2 Sample preparation

The setup was first calibrated using a silicon sample that was first crushed to a fine powder, and then added to a capillary tube which was sealed by heat. The crystal was crushed into a fine powder to minimize preferred orientation effects and ensure a random distribution of orientations. The tube was then inserted into a goniometer. The goniometer was used to align the capillary tube with the source beam. After alignment, the goniometer was placed on the stage, where the sample was examined using the diffraction setup as shown in Figure 1. This process was repeated in order to characterize the unknown sample which could be described as a white powder.

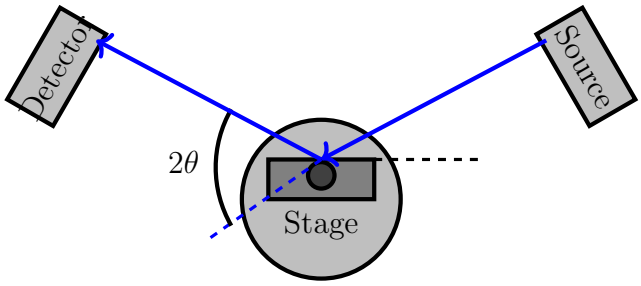

Figure 1. A schematic representation of the X-ray diffraction setup. A sample of specimen is prepared and placed on the sample stage. A monochromatic X-ray source is used for diffraction. The diffracted beams are captured by an X-ray detector.

4 Results

4.1 The Si diffraction

The diffractogram of the silicon reference sample is shown in Figure 2a. In Table 1 we have summarized the first three experimental peak characteristics together with the three lowest theoretical reflections. We see that there is a good correspondence between the theoretical values and the experimental values on the first three peaks, corresponding to the planes $(1\,1\,1), (2\,2\,0)$ and $(3\,1\,1)$. We also estimated the lattice constant a using (2) on the experimental d-spacing. These values are also shown in Table 1.

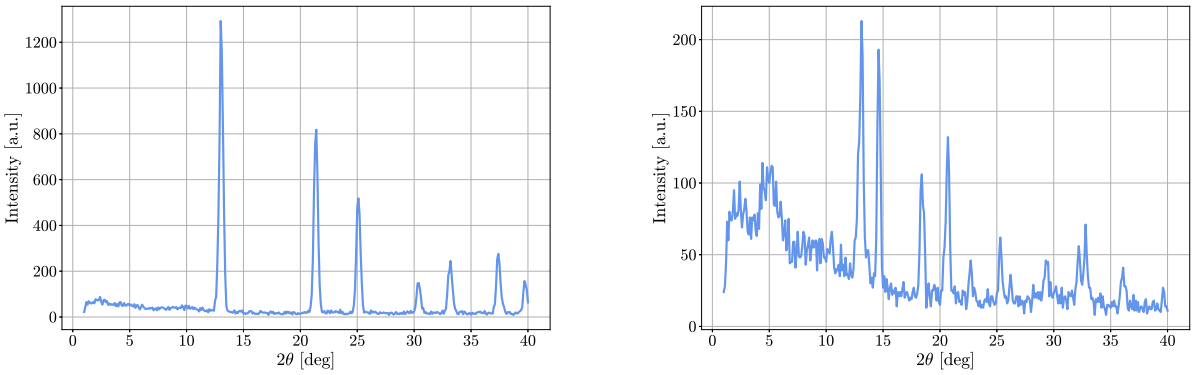

Figure 2. Experimental diffractogram of the diffracted intensity I as a function of the angular position 2θ for a) Si and b) the unknown sample. Note the distinct peaks of the Si specimen, and the peak pairs in the unknown specimen.

Table 1. (Left) Experimental values for angular positions, d-spacings, intensities and lattice constants for the first three peaks of Si, and (Right) theoretical values for Miller planes, angular positions and d-spacings of the three lowest reflections in Si.

Peak	2θ [°]	d_{hkl} [Å]	I [a.u]	a [Å]	$(h\,k\,l)$	2θ [°]	d_{hkl} [Å]
1	13.0	3.133	1293	5.427	$(1\,1\,1)$	12.99	$\frac{a}{\sqrt{3}}$
2	21.4	1.910	818	5.402	$(2\,2\,0)$	21.29	$\frac{a}{\sqrt{8}}$
3	25.1	1.630	518	5.406	$(3\,1\,1)$	25.02	$\frac{a}{\sqrt{11}}$

The theoretical values of 2θ gives us the average error $\Delta 2\theta = 0.07°$. Using Gauss propagation law on the lattice constant as a function of 2θ, we calculate the average value of the lattice constant as $\bar{a} = 5.412(\pm 0.014)$ Å.

4.2 The unknown diffraction

The diffractogram of the unknown sample is illustrated in Figure 2b. The sample substance was observed by eye to be white. We calibrated the angle measurements to the values from the silicon sample. The calibrated angles, intensity and calculated d-spacing for the first six peaks are given in Table 2. Utilizing the Webmineral mineralogy database [4] we found no suitable candidate for the sample. There were multiple substances with the right d-spacings corresponding to the first three peaks, but with wrong color and the wrong relative intensities at these peaks.

The observation that the intensity in the peaks do not drop of monotonically indicates that the sample might include more than one substance. The peaks also appear to be placed in pairs for the first four peaks. We proceed with the hypotheses that the sample contains two substances, with peaks alternating as function of 2θ. Under this assumption, we split up the peaks in our data.

Table 2. Experimental values for angular positions, d-spacings and intensities for the first six peaks of the unknown sample, using angles calibrated to the silicon sample

Peak	1	2	3	4	5	6
2θ [°]	13.0	14.5	18.3	20.6	22.6	25.2
d_{hkl} [Å]	3.133	2.810	2.230	1.983	1.810	1.626
I [a.u]	213	193	106	122	46	56

Utilizing webmineral [4], we found possible candidates that matches the values in Table 2 with a 5% tolerance that also match the intensity distribution. For the first grouping, peaks 1,3 and 5, the possible candidates are RhAs and KCl. For the second grouping, peaks 2, 4 and 6, the possible candidates are NaCl, (Ca, Mg, Fe)S and AgBr. RhAs, (Ca, Mg, Fe)S, and AgBr can be discarded as these materials are known to be black, brown, and yellow respectively [4]. As such, the two remaining candidates are KCl and NaCl, as both are white.

The simplicity and abundance of these materials make them highly believable candidates.

4.3 Sources of error

The calculated average lattice constant of silicon, agrees well with the theoretical value of 5.4309 Å. This is within the margin of error from the uncertainty in 2θ.

The uncertainty in 2θ for silicon is reasonable as well, however as all experimental values are larger than the theoretical ones, there might be some systematic source of error here. The minor deviations in the 2θ-positions may likely be due to a combination of instrumental misalignment and wavelength spread from the graphite monochromator.

5 Conclusion

In the sample known to contain silicon, the diffractogram gave peaks in intensity given by Table 1. These fit well with known peaks for silicon as well as those theoretically predicted. The lattice constant calculated from these also fit with the theoretical values. The unknown sample peaks matched with a combination of KCl and NaCl. The major sources of error could be due to possible instrumental misalignments and the source of beam. These were not detrimental to the experiment.

References

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